Indigenous Knowledge and Geospatial Bias in the Era of Large Language Models

ACM SIGSPATIAL 2026 TUTORIAL

Nicole R. Schneider | Nandini Ramachandran | Kent O'Sullivan | Hanan Samet

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Acknowledgement of Country

- We respectfully acknowledge and recognize our responsibility to the original and current caretakers of the land, water, and air where this conference takes place: the **Cahuilla, Tongva, Luiseño, and Serrano** peoples, and all of their ancestors and descendants, past, present, and future.

- We pay respects to Elders past and present, and to the **Aboriginal and Torres Strait Islander, First Nations, and Māori** communities whose place names and knowledge appear in this tutorial.

- Some content discusses culturally sensitive knowledge. We present it with care and with reference to community-led governance principles.

Tutorial Roadmap (90 minutes)

Time	Section	Highlights
0–10 min	Introduction & Motivation	What is geospatial bias? Value of Indigenous knowledge. Why representation in AI matters.
10–25 min	Part I: Foundations of Indigenous Principles	Indigenous Data Sovereignty, CARE, Indigenous data sources, truth-telling.
25–45 min	Part II: AI Fairness and Open Data	FAIR & fairness metrics; debiasing; embedding geometry & unlearning; retrieval governance.
45–50 min	Break / Interactive Discussion	Audience discussion prompts.
50–65 min	Part III: Case Study	Dual-named places; Indigenous place names in embedding space; fine-tuning.
65–75 min	Part IV: Representation in Tech	Workforce under-representation ↔ geo-bias; organizational efforts.
75–90 min	Challenges, Q&A, Closing	Open research questions and discussion.

What You Will Take Away

- A working definition of **geospatial bias** in LLMs and why Indigenous knowledge is a prime example of it.
- The core **Indigenous data principles** (Indigenous Data Sovereignty, CARE) and how they complement **FAIR**.
- A map of **fairness techniques**: evaluation metrics, debiasing, embedding-space analysis, machine unlearning.
- Why **retrieval architecture is a governance decision** (dense vector RAG vs. knowledge graphs / SQL).
- Hands-on intuition from a **case study** of Indigenous vs. Anglicized place names in LLM embedding space.
- A set of **open research questions** for the SIGSPATIAL community.

Who Is This Tutorial For?

- Researchers and practitioners interested in:
  - AI and geospatial fairness
  - Indigenous knowledge and practices
  - Language and place representation in LLMs
  - Minority representation in technology
- **Prerequisites:** none required; basic familiarity with AI and geospatial concepts helps.
- **Materials provided:** Indigenous place name dataset, Python notebook to visualize embeddings, slides, and a summary of organizations advancing Indigenous representation.

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Three-circle Venn diagram: Geospatial, AI Fairness, and Indigenous Knowledge.

Introduction & Motivation

0–10 min

Introduction & Motivation (0–10 min)

Part I: Foundations of Indigenous Principles (10–25 min)

Part II: AI Fairness and Open Data (25–45 min)

Break / Interactive Discussion (45–50 min)

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Part IV: Indigenous Representation in Tech (65–75 min)

Challenges, Q&A, and Closing (75–90 min)

LLMs Are Becoming Geospatial Search Engines

- LLMs like ChatGPT and Claude are increasingly used for tasks with a geospatial component:
  - Planning travel itineraries and routes
  - Retrieving historical information about places
  - Finding places and points of interest
  - Question answering and document editing
- As LLMs replace traditional search and map interfaces, they reshape **whose knowledge is treated as authoritative** and what information is accessible.

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Mock chat screenshot highlighting Anglicized vs. Indigenous place names.

What Is Geospatial Bias?

- Systematic differences in the **quality, accuracy, or representation** of model outputs across places, peoples, languages, and world views.
- Root causes:
  - LLMs are trained by a small number of resource-rich organizations
  - Internet-scale training data under-represents less popular and out-of-distribution information
  - Result: models reflect certain places and world views and perform poorly where information is not prevalent online
- The inequity appears **not only in outputs, but in the model's internal latent space** — output-level bias audits alone cannot fix it.

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World map comparing web data density with LLM place accuracy.

Motivating Example: One Place, Two Names

- Many places have both an **Indigenous** and an **Anglicized** name (dual naming):
  - Uluru / Ayers Rock (Australia)
  - Aoraki / Mount Cook (New Zealand)
  - Haida Gwaii / Queen Charlotte Islands (Canada)
- Ask an LLM the **same factual question** (e.g., population of a town) with each name → answers differ [17].
- Same place, same fact — different model behavior.

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Side-by-side chat responses for dual place names with ground truth.

The Value of Indigenous Knowledge

- Indigenous knowledge is often **passed down orally** rather than written or digitized [16].
- Place names encode history, ecology, navigation, kinship, and law — they are knowledge, not just labels.
- Indigenous value systems tend to be **community-centric** and deeply tied to land and customs [2, 8].
- Many Indigenous Peoples hold **Sacred Knowledge** that is not meant to be shared freely.
- ⇒ Improving Indigenous representation in LLMs is not purely technical; it is a matter of **information ownership and governance**.

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Diagram unpacking an Indigenous toponym's meaning and cultural context.

Two Axes of Fairness

- **Representational fairness:** do outputs and internal representations treat Indigenous and non-Indigenous information equivalently?
- **Data & algorithmic sovereignty:** do communities retain authority over how their knowledge is collected, stored, retrieved, and used?
- A model can score well on the first axis while failing the second — e.g., equal accuracy across Anglicized and Indigenous names achieved by embedding culturally sensitive information **without consent**.
- The sovereignty axis is under-explored in a literature that defaults to **fairness-as-parity**.

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2×2 quadrant: representational parity vs. community sovereignty.

Part I: Foundations of Indigenous Principles

10–25 min

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Indigenous Data Sovereignty (IDS)

- Indigenous Peoples have the right to control the **collection, ownership, governance, and application** of data about their communities, lands, and resources [18].
- IDS provides a framework for maximizing the benefit of open data for Indigenous Peoples **and** other users of Indigenous data.
- Simply adding more Indigenous data to training corpora risks further misrepresentation if done without community consent and governance.
- **Technical inclusion ≠ ethical inclusion.**

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Data lifecycle loop centering Indigenous community governance.

The CARE Principles for Indigenous Data Governance [4]

C — Collective Benefit	A — Authority to Control	R — Responsibility	E — Ethics
Data ecosystems should be designed so Indigenous Peoples derive benefit from the data.	Indigenous Peoples' rights and interests in Indigenous data must be recognized; they decide how it is governed.	Those working with Indigenous data must show how its use supports Indigenous self-determination and collective benefit.	Indigenous Peoples' rights and wellbeing should be the primary concern at all stages of the data life cycle.

Developed (Carroll et al., 2020) to complement FAIR: centers people, power, and purpose — not just data.

FAIR and CARE: Complementary, Not Competing

	FAIR [22]	CARE [4]
Focus	Data-centric: technical characteristics	People- and purpose-centric
Goal	Increase data sharing & machine reuse	Indigenous rights, collective benefit, governance authority
Principles	Findable, Accessible, Interoperable, Reusable	Collective Benefit, Authority to Control, Responsibility, Ethics
Blind spots	Power differentials, historical context, authority over data	Not a technical spec — must be operationalized in systems
In LLM systems	Open corpora, open embeddings, open benchmarks	Consent, provenance, revocation, community review

"Be FAIR and CARE" (GIDA): systems for Indigenous data should satisfy both.

Indigenous Datasets and Data Sources

- Official gazetteers with dual / Indigenous names [candidates — confirm]:
  - Australia: state geographic names boards; AIATSIS language resources
  - Aotearoa New Zealand: NZ Gazetteer (LINZ / NZ Geographic Board)
  - Canada: Canadian Geographical Names Database (CGNDB)
- Community-led mapping initiatives (e.g., Native Land Digital) — note their own usage caveats.
- Our tutorial dataset: Indigenous ↔ Anglicized place name pairs with locations (released to attendees).
- Every dataset raises governance questions: **who collected it, with whose consent, under what terms?**

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Dataset cards comparing FAIR-only vs. CARE-aware governance.

Truth-Telling of Indigenous History

- A growing movement toward **truth-telling** about colonial history and Indigenous Peoples' experiences.
- Examples to highlight [verify dates / wording]:
  - Canada: Truth and Reconciliation Commission (final report 2015)
  - Australia: Uluru Statement from the Heart (2017) calls for truth-telling
  - Place name restoration and dual naming in Australia, Canada, and New Zealand
- Implication for AI: models trained on historical web text may **encode outdated or colonial narratives** about places.

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Timeline of place-name restoration and truth-telling milestones.

Part II: AI Fairness and Open Data

25–45 min

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Part II at a Glance

1. Frameworks & Evaluation	2. Model Debiasing	3. Vector & Geometry	4. Retrieval Governance
FAIR principles; standard fairness metrics applied to geography.	Causality-guided [15]; agent-based [9]; data augmentation & fine-tuning [11].	Latent embedding space analysis [7, 10, 14]; machine unlearning [3, 13].	Dense vector RAG [19] vs. knowledge graphs & relational storage [21].

Thread throughout: how does each technique serve representational fairness AND sovereignty?

Evaluating Fairness: Metrics Applied to Geography

- Standard metrics, adapted to places:
  - **Accuracy gap:** Δ = acc(Anglicized name) − acc(Indigenous name)
  - **Counterfactual consistency:** swap the toponym, keep the question — does the answer change?
  - **Group parity:** error / refusal / hallucination rates across regions or language groups
  - **Retrieval exposure:** how often Indigenous sources are retrieved / ranked / cited
- IR & LLM bias taxonomies: source bias, popularity bias, unfairness in ranking and generation [5, 6].
- Limitation: metrics measure **parity**, not **sovereignty** (recall the two axes).

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Bar chart comparing LLM accuracy across name types with gap Δ.

Model Debiasing: Three Families

Approach	Key idea	Considerations for Indigenous knowledge
Causality-guided [15]	Model how sensitive attributes causally affect outputs; steer the LLM (e.g., via prompting) away from biased pathways.	No retraining needed; requires a causal model of what ‘bias’ means for places and names.
Agent-based [9]	Multiple agents debate and interpret token-level bias (e.g., UMAD) before answering.	Inference-time cost; agents share the same knowledge gaps if facts are absent.
Data augmentation & fine-tuning [11]	Generate counterfactual / balanced data (e.g., with ChatGPT) and apply parameter-efficient fine-tuning.	Most direct fix for missing knowledge — but raises consent & sovereignty questions over the data used.

Vector & Geometry: Inside the Embedding Space

- Contextual embedding spaces are **anisotropic**: less frequent / lower-resource terms occupy narrower, more distorted regions [7].
- A bottleneck occurs even earlier: non-alphabetic Indigenous scripts face **substantial tokenization costs** before embedding [14].
- LLMs encode geographic coordinates as **linear directions** for well-documented places [10].
- Open question: do Indigenous toponyms receive comparable linear structure?

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Anisotropic embedding space schematic showing place name clustering.

Machine Unlearning

- Goal: **selectively remove the influence of specific data** from an already-trained model [3, 13].
- Approaches:
  - Exact: sharded training (SISA) — retrain only affected shards [3]
  - Approximate: knowledge unlearning via targeted gradient updates [13]
- Why it matters here: a technical mechanism for **Authority to Control** — communities could revoke knowledge after the fact.
- Challenges: verification of forgetting, cost, collateral damage to related knowledge.

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Sharded model diagram illustrating selective unlearning upon consent revocation.

Retrieval Architecture Is a Governance Decision

	Dense vector RAG [19]	Knowledge graph / SQL [21]
Representation	Text compressed into fixed-length vectors	Explicit records / triples, never collapsed into a vector
Provenance & consent metadata	Discarded by construction once embedded	Preserved as explicit links (source, permission, restriction)
Revocation	No principled way to trace or selectively revoke	Delete / restrict a record; enforce at query time
Matching	Fuzzy semantic similarity	Exact queries (LLM writes SPARQL / SQL)
Reasoning (DistRAG findings)	Handles ~1-hop; fails multi-hop	Perfect up to 1-hop; breaks on multi-hop

Dense Vector RAG: Where Provenance Gets Lost

- Pipeline (DistRAG [19]): knowledge triples → text → embedding model → FAISS vector store → top-k retrieval → LLM.
- Once a passage is embedded, there is no principled way to trace it back to community authorization.
- Similarity search can surface semantically ‘close’ but culturally inappropriate content.

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DistRAG flow highlighting where provenance metadata is lost in encoding.

Structured Retrieval: Knowledge Graphs & SQL

- LLM translates the question into a **SPARQL / SQL** query; the engine executes it on a structured store [19, 21].
- Each fact can carry **source, permission, and access-restriction** attributes.
- Access policies can be enforced **at the retrieval layer**, before anything reaches the LLM.
- Trade-offs: schema design effort, brittleness of LLM-generated queries, multi-hop reasoning.

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Knowledge graph snippet with provenance and access restriction filters.

Break / Interactive Discussion

- Should an LLM answer questions about a sacred site at all? Who decides?
- If a place has two official names, which one should a model use by default?
- Is equal accuracy across name types a sufficient fairness goal?
- Would you trade retrieval accuracy for enforceable consent metadata?

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Interactive poll QR code for audience responses.

Part III: Case Study — Indigenous Place Names

50–65 min

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Dual Naming in Australia, Canada, and New Zealand

- **Dual naming:** officially recognizing both an Indigenous and an introduced name for the same place [12].
- History: language planning and place-naming policy in Australia formalized dual naming from the 1990s [12].
- Examples [verify]: Uluru / Ayers Rock; Kati Thanda–Lake Eyre; Aoraki / Mount Cook; Haida Gwaii.
- Dual names give a natural **controlled experiment**: same entity, same facts, two surface forms.

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Map of dual-named places across Australia, Canada, and New Zealand.

Experimental Setup

- Task: **population estimates** for Australian locations with dual names [17].
- For each place: query the LLM with the Indigenous toponym and with the Anglicized toponym.
- Measure: response accuracy vs. ground truth, and consistency between the two names.
- Extract internal embeddings for both names and compute **dispersion** of their representations.

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Evaluation pipeline: prompt template, LLM answers, and embedding dispersion.

Finding: Accuracy Tracks Embedding Dispersion

- LLMs respond differently to factual queries depending on whether the Indigenous or Anglicized name is used [17].
- Prior work establishes a **correlation between response accuracy and the dispersion** of the internal representation of Indigenous toponyms [17].
- Ongoing work shows a **representational gap** for Indigenous place names in current LLMs.
- ⇒ Fixing outputs is not enough; **embeddings must improve** to reduce geospatial bias.

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Scatter plot: embedding dispersion vs. population estimation error.

Visual Exploration: Place Names in Embedding Space

- Project embeddings to 2-D (PCA / t-SNE / UMAP).
- Color by name type; draw a line between each Indigenous ↔ Anglicized pair.
- Look for: clusters by language, distance between synonym pairs, collapse of Indigenous names into a narrow region.
- Related: cross-lingual clustering with LLMs [20].

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2-D projection comparing Indigenous and Anglicized embedding clusters.

The Indigenous Place Name Dataset

- Released to attendees with Python loading & visualization code.
- Proposed schema [draft]:
  - Indigenous name, Anglicized name, language group
  - Coordinates, state / region, place type
  - Ground-truth attributes (e.g., population)
  - Source / provenance and usage notes
- Governance: document how names were sourced and any community usage terms (CARE).

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Dataset sample table with regional coverage map.

Fine-Tuning LLMs on Indigenous Place Names

- Idea: fine-tune (e.g., parameter-efficient / LoRA) on curated Indigenous place-name data.
- Measure: change in accuracy gap Δ and in embedding geometry for target terms.
- Questions: how many examples are needed? Does it transfer across models? Does it hurt other knowledge?
- Must be **community-curated and consented** — otherwise it fails the sovereignty axis.

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Embedding plots showing cluster alignment before and after fine-tuning.

Hands-On: Tutorial Materials

- Notebook walkthrough:
  - 1. Load the Indigenous place name dataset
  - 2. Embed names with an open LLM / embedding model
  - 3. Visualize pairs in 2-D and compute dispersion
  - 4. Run counterfactual queries and compute Δ
- Link: [GitHub / Colab URL — TBD]

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Repository QR code and tutorial notebook preview.

Part IV: Indigenous Representation in Tech

65–75 min

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Who Builds AI Shapes What AI Knows

- Lack of Indigenous representation in the research and industry fields that build models and curate training data contributes to under-representation in AI.
- Decisions shaped by who is in the room: which data is collected, which benchmarks matter, which errors get noticed.
- Geospatial bias in models is therefore a **holistic, socio-technical** issue.

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Feedback loop: AI workforce, data curation, benchmarks, and community impact.

Organizational Efforts to Increase Representation

IWGIA	IITC	CSIRO Indigenous Graduate Program	More to add [optional]
International Work Group for Indigenous Affairs — Indigenous rights research, documentation, and advocacy.	International Indian Treaty Council — Indigenous-led organization advocating for Indigenous Peoples' rights.	Australia's national science agency connecting Indigenous graduates with STEM research careers.	e.g., AISES; Indigenous Protocol and AI Working Group; Te Hiku Media (Māori data sovereignty); Local Contexts TK / BC Labels.

Materials will include a summary of organizations with initiatives to advance Indigenous representation.

Challenges, Q&A, and Closing

75–90 min

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Open Questions: Representation & Geometry

- Which embedding-space properties — **dispersion, anisotropy, neighborhood density** — best predict accuracy loss for under-represented toponyms?
- Do these properties **transfer across models**?
- Do Indigenous toponyms receive linear geographic structure comparable to well-documented places [10]?
- Can **community-curated datasets** shift embedding geometry for target terms without full retraining — and how many examples would this require?

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Discussion icon grid representing key open research questions.

Open Questions: Retrieval & Governance

- Can retrieval architecture (KG / SQL vs. dense RAG) **close representational gaps without retraining**, and at what cost?
- How can **provenance and consent metadata** be enforced at the retrieval layer?
- Can **unlearning, retrieval-layer filtering, or knowledge editing** let communities revoke specific knowledge at inference time?
- How do we evaluate the **sovereignty axis**, not just parity?

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Updated fairness quadrant mapping proposed research directions.

Key Takeaways

- Geospatial bias in LLMs is real, measurable, and lives in the **latent space**, not just outputs.
- Indigenous knowledge is a prime case: oral, community-centric, under-represented online.
- Fairness has **two axes**: representational parity and data / algorithmic sovereignty.
- Technical methods (evaluation, debiasing, fine-tuning, unlearning) must be paired with governance (IDS, CARE).
- **Retrieval architecture is a governance decision.**

Resources

- Tutorial website: [URL — TBD]
- Dataset + Python notebook: [GitHub / Colab — TBD]
- Slides: [link — TBD]
- Organizations advancing Indigenous representation: summary handout
- Related tutorials: WSDM'25 "Unifying Bias and Unfairness in IR"; ACL'25 "Navigating Ethical Challenges in NLP" [1, 5, 6]
- Our related work: DistRAG [19]; Embeddings Matter [17]; Cross-Lingual Clustering [20]; Engineering Prompts for Spatial Questions [21]

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Tutorial website QR code and resources link.



Questions

Nicole Schneider nsch@umd.edu

Nandini Ramachandran nandinir@terpmail.umd.edu

Kent O'Sullivan kosu0918@uni.sydney.edu.au

Hanan Samet hjs@cs.umd.edu

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